Data Exploration – Bivariate Analysis

CS 501
Advanced Topics in Computer Science
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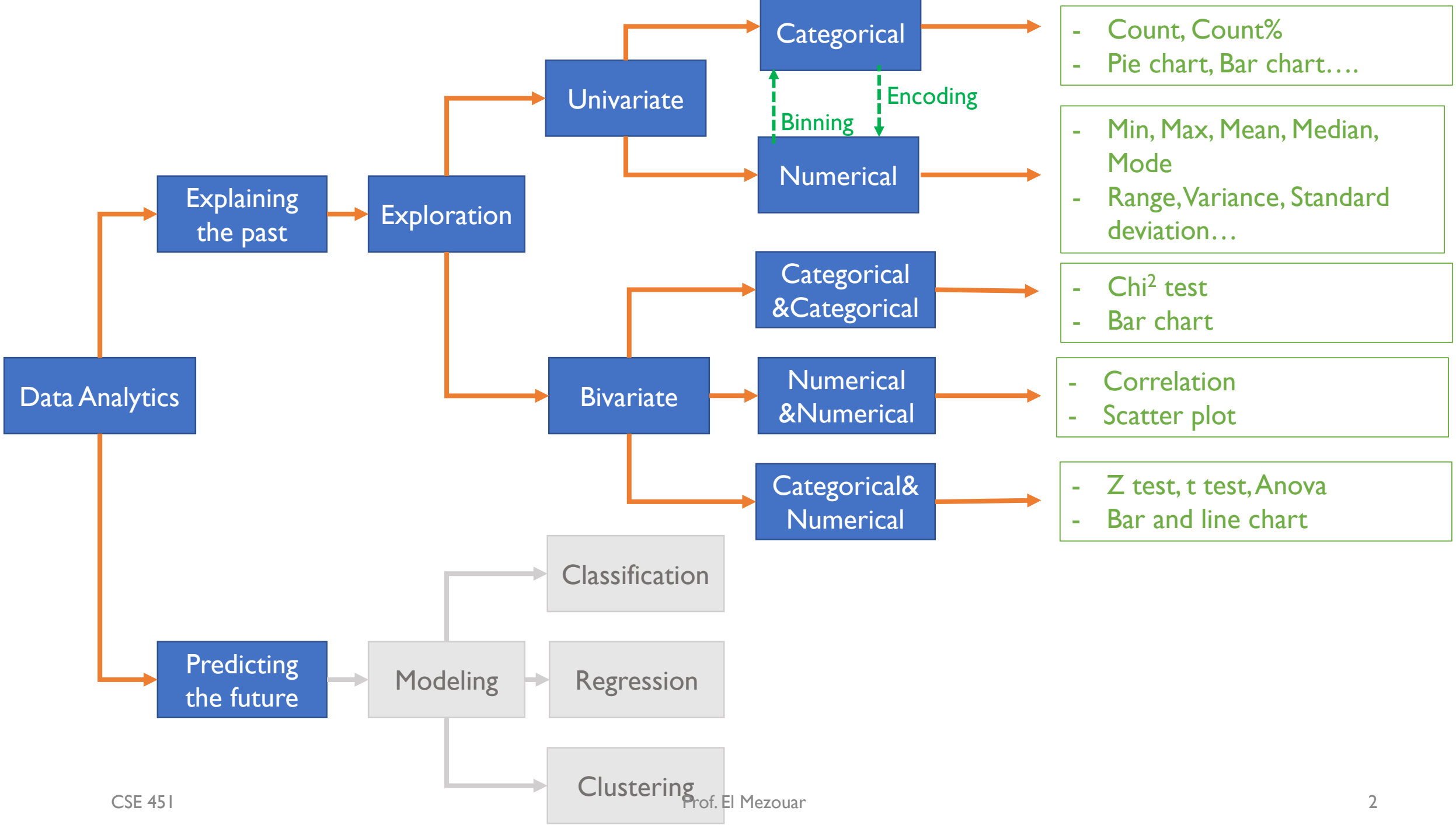
Data Analytics
Explaining the past
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- Count, Count%
- Pie chart, Bar chart….
- Min, Max, Mean, Median, Mode
- Range, Variance, Standard deviation…
- Chi2 test
- Bar chart
- Correlation
- Scatter plot
- Z test, t test, Anova
- Bar and line chart

BIVARIATE ANALYSIS

Introduction

- Bivariate analysis is the simultaneous analysis of two variables (attributes).

- It explores the concept of relationship between two variables,
  - whether there *exists an association* and the *strength* of this association, or
  - whether there are *differences between two variables* and the *significance* of these differences.

Introduction

- There are three types of bivariate analysis

    1. Numerical & Numerical
    2. Categorical & Categorical
    3. Numerical & Categorical

Numerical and Numerical

Scatter Plot

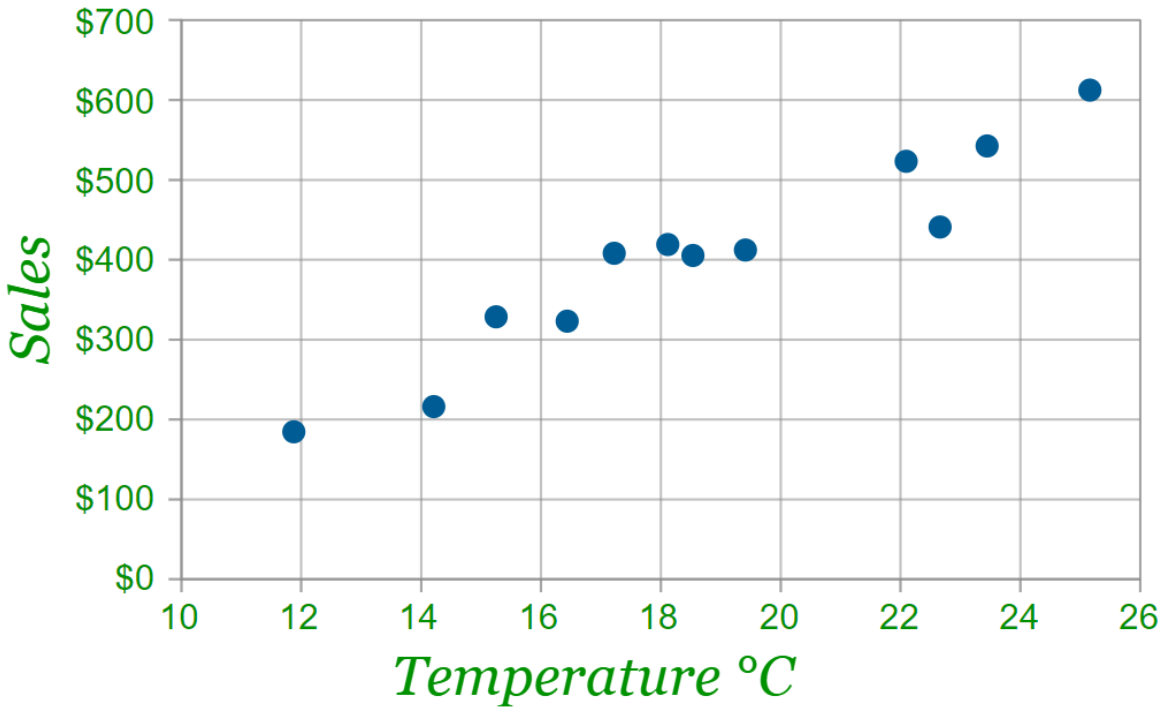
- A scatter plot is a useful visual representation of the relationship between two numerical variables (attributes).
- It is usually drawn before working out a linear correlation (?) or fitting a regression line (?).

- The resulting pattern indicates the type (linear or non-linear) and strength of the relationship between two variables.

Example

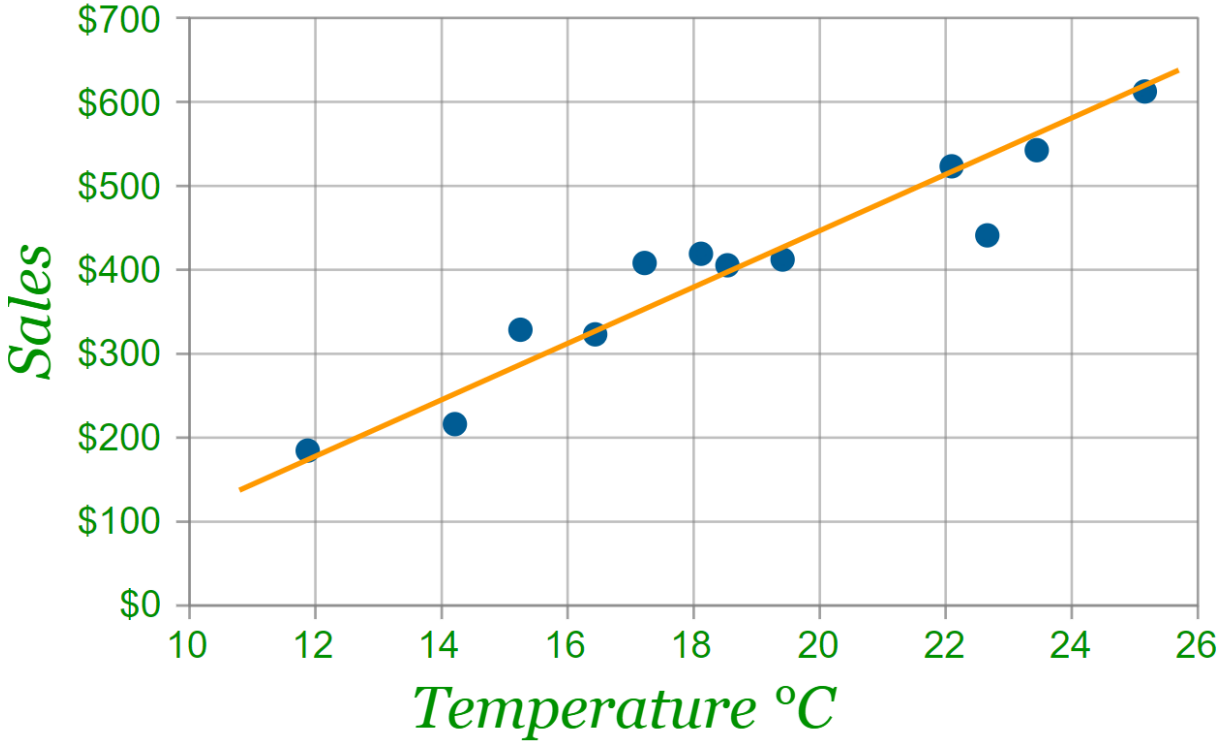
Ice Cream Sales vs Temperature

Temperature °C	Ice Cream Sales
14.2°	$215
16.4°	$325
11.9°	$185
15.2°	$332
18.5°	$406
22.1°	$522
19.4°	$412
25.1°	$614
23.4°	$544
18.1°	$421
22.6°	$445
17.2°	$408



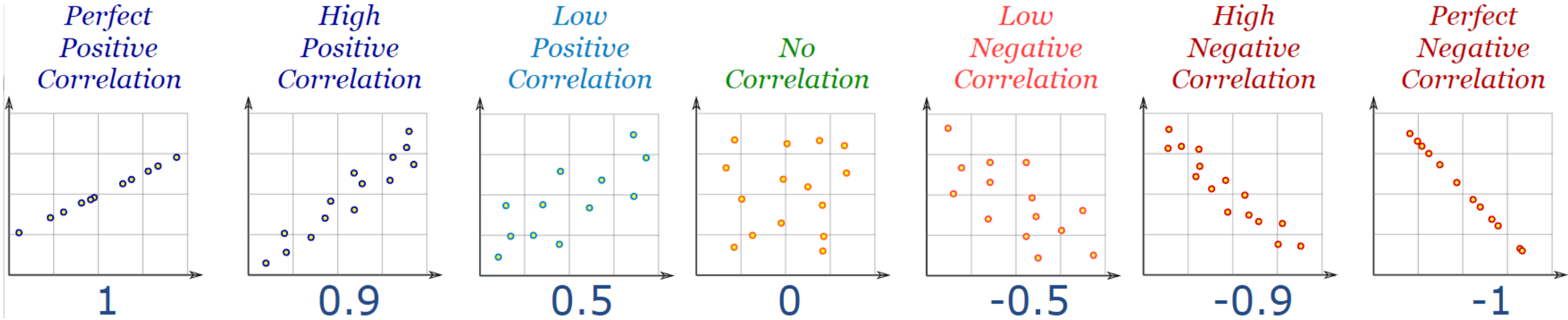
Example - Cont.,

- We can also draw a "Line of Best Fit" (also called a "Trend Line") on our scatter plot:
  - Try to have the line **as close as possible to all points**, and as many points above the line as below.



Correlation

- When the two sets of data are strongly linked together we say they have a **High Correlation**.
  - Correlation is **Positive** when the values **increase** together, and
  - Correlation is **Negative** when one value **decreases** as the other increases



Correlation – Cont.,

Correlation can have a value:

- 1 is a perfect positive correlation
- 0 is no correlation (the values don't seem linked at all)
- -1 is a perfect negative correlation

<u>Formally speaking</u>: (Linear) correlation quantifies the strength of a linear relationship between two numerical variables.

- When there is no correlation between two variables, there is no tendency for the values of one quantity to increase or decrease with the values of the second quantity.

Correlation

$$r = \frac{Covar(x,y)}{\sqrt{Var(x)Var(y)}}$$

$$Covar(x,y) = \frac{\sum(x-\bar{x})(y-\bar{y})}{n}$$

$$Var(x) = \frac{\sum(x-\bar{x})^2}{n}$$

$$Var(y) = \frac{\sum(y-\bar{y})^2}{n}$$

r : *Linear Correlation*

$Covar$: *Covariance*

Var : *Variance*

r only measures the **<u>strength of a linear relationship</u>** and is always *between -1 and 1* where

- -1 means perfect negative linear correlation and
- +1 means perfect positive linear correlation and
- 0 means no linear correlation.

Example

Temperature	83	64	72	81	70	68	65	75	71	85	80	72	69	75
Humidity	86	65	90	75	96	80	70	80	91	85	90	95	70	70

	Variance	Covariance	Correlation
Temperature	40.10	19.78	0.32
Humidity	98.23		

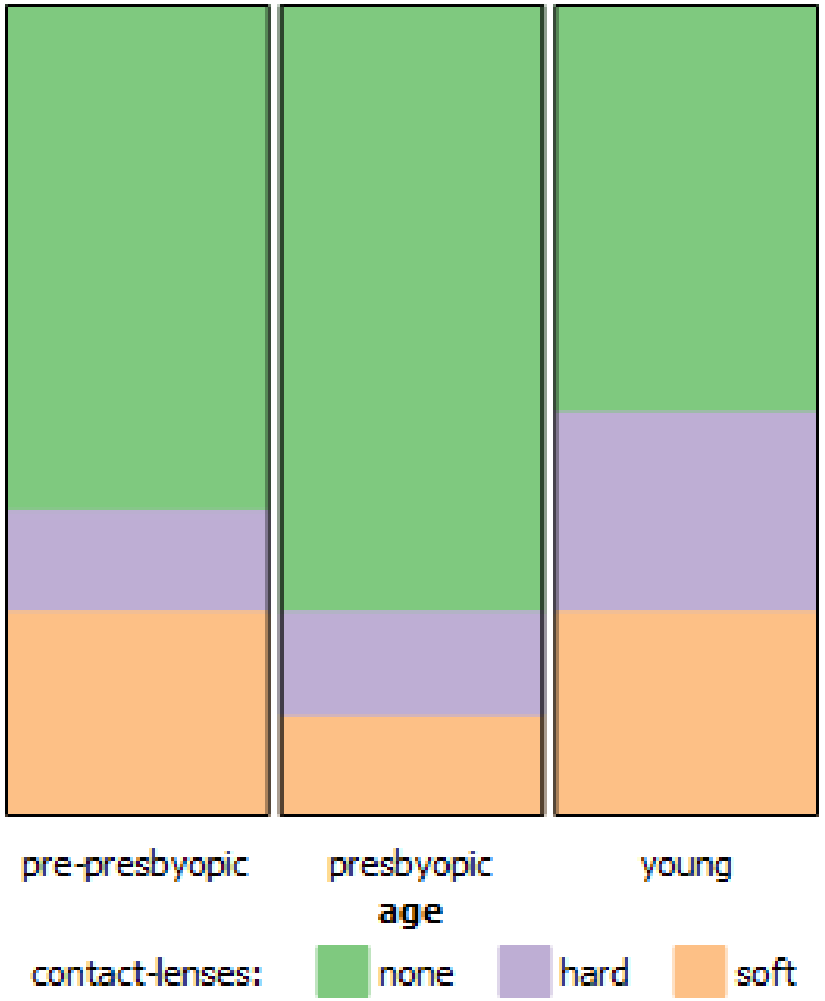
There is a *weak linear correlation* between Temperature and Humidity.

Categorical & Categorical

Stacked Column Chart

- Stacked Column chart is a useful graph to visualize the relationship between two categorical variables.

- It compares the percentage that each category from one variable contributes to a total across categories of the second variable.



Chi-square Test for Independence

- The chi-square test can be used to determine the association between categorical variables.

- It is based on the difference between the expected frequencies (*e*) and the observed frequencies (*n*) in one or more categories in the frequency table.

Chi-square Test for Independence

- *The chi-square distribution* returns a probability for the computed chi-square and the degree of freedom.
- A probability of zero shows a complete dependency between two categorical variables and a probability of one means that two categorical variables are completely independent.
  - *Tchouproff Contingency Coefficient* measures the amount of dependency between two categorical variables.

Chi-square Test

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(n_{ij} - e_{ij})^2}{e_{ij}}$$

$$e_{ij} = \frac{n_{i.} n_{.j}}{n}$$

Tchouproff Contingency Coefficient

$$\rho_\tau = \sqrt{\frac{\chi^2}{n\sqrt{(c-1)(r-1)}}}$$

e - the expected frequencies
n - the observed frequencies
r – number of rows in contingency table
c – number of columns in contingency table

Chi-square Test

e=(44 * 52)/95=24.1

Hair

Eye	Light	Dark	
Black	**32** (24.1)	**12** (19.9)	44
Green/Blue	**14** (19.7)	**22** (16.3)	36
Others	**6** (8.2)	**9** (6.8)	15
	52	43	95

$$\chi^2 = 10.67$$

$$\rho_\tau = \sqrt{\frac{10.67}{95\sqrt{(3-1)(2-1)}}} = 0.28$$

Categorical & Numerical

Line Chart with Error Bars

- A line chart with error bars displays information as a series of data points connected by straight line segments.
- Each data point is average of the numerical data for the corresponding category of the categorical variable with error bar showing standard error SE.
- It is a way to summarize how pieces of information are related and how they vary depending on one another

$$\mathrm{SE} = \frac{\sigma}{\sqrt{n}}$$

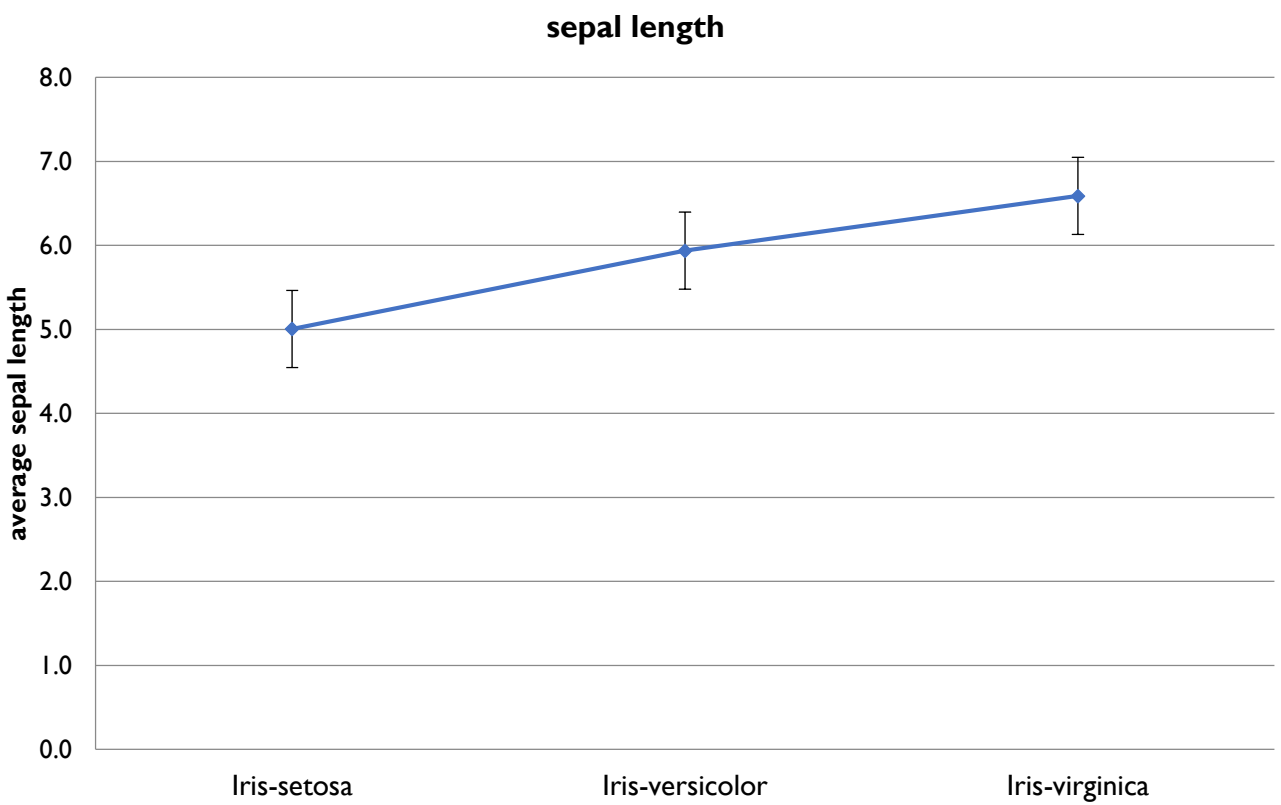
SE = standard error of the sample

σ = sample standard deviation

n = number of samples

Line Chart with Error Bars

sepal length		
Iris-setosa	Iris-versicolor	Iris-virginica
5.1	7	6.3
4.9	6.4	5.8
4.7	6.9	7.1
4.6	5.5	6.3
5	6.5	6.5
5.4	5.7	7.6
4.6	6.3	4.9
5	4.9	7.3
4.4	6.6	6.7
4.9	5.2	7.2
...	...	...



Z-test and t-test

- Z-test and t-test are basically the same.

- They assess whether the averages of two groups are statistically different from each other.

- This analysis is appropriate for comparing the averages of a numerical variable for two categories of a categorical variable.

Z-test

$$Z = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{\frac{S_1^2}{N_1} + \frac{S_2^2}{N_2}}}$$

where:

- $\overline{X}_1, \overline{X}_2$: *Averages*
- S_1^2, S_2^2: *Variances*
- N_1, N_2: *Counts*
- Z: *Standard Normal Distribution*

If the probability of Z is small, the difference between two averages is more significant.

T-test

- When the n_1 or n_2 is less than 30 we use the t-test instead of the Z-test

$$t = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{S^2\left(\frac{1}{N_1} + \frac{1}{N_2}\right)}}$$

$$S^2 = \frac{(N_1 - 1)S_1^2 + (N_2 - 1)S_2^2}{N_1 + N_2 - 2}$$

where:

- $\overline{X}_1, \overline{X}_2$: *Averages*
- S_1^2, S_2^2 : *Variances*
- N_1, N_2 : *Counts*
- t : has t distribution with $N_1 + N_2 - 2$ degree of freedom

Example

- Is there a significant difference between the means (averages) of the numerical variable (Temperature) in two different categories of the categorical variable (O-Ring Failure)?

O-Ring Failure	Temperature
Y	53 56 57 70 70 70 75
N	63 66 67 67 67 68 69 70 72 73 75 76 76 78 79 80 81

t-test	O-Ring Failure	
Temperature	Y	N
Count	7	17
Mean	64.43	72.18
Variance	76.95	30.78
t	-2.62	
df	22	
Probability	0.0156	

Example

t-test	**O-Ring Failure**	
Temperature	Y	N
Count	7	17
Mean	64.43	72.18
Variance	76.95	30.78
t	-2.62	
df	22	
Probability	0.0156	

The low probability (0.0156) means that the difference between the average temperature for failed O-Ring and the average temperature for intact O-Ring is significant

Analysis of Variance (ANOVA)

- The ANOVA test assesses whether the averages of more than two groups are statistically different from each other.
- This analysis is appropriate for comparing the averages of a numerical variable for more than two categories of a categorical variable.

Source of Variation	Sum of Squares	Degree of Freedom	Mean Square	F	P
Between Groups	SS_B	df_B	$MS_B = SS_B/df_B$	$F=MS_B/MS_W$	P(F)
Within Groups	SS_W	df_w	$MS_W = SS_W/df_w$		
Total	SS_T	df_T			

Analysis of Variance (ANOVA)

$$SS_B = \sum_{i=1}^{k} \frac{(\sum X)_i^2}{N_i} - \frac{\left(\sum_{i=1}^{k} (\sum X)_i\right)^2}{\sum_{i=1}^{k} N_i}$$

$$df_W = \sum N_i - k$$

$$df_B = k - 1$$

$$SS_W = \sum_{i=1}^{k} (\sum X^2)_i - \sum_{i=1}^{k} \frac{(\sum X)_i^2}{N_i}$$

$$df_T = \sum N_i - 1$$

$$SS_T = \sum_{i=1}^{k} (\sum X^2)_i - \frac{\left(\sum_{i=1}^{k} (\sum X)_i\right)^2}{\sum_{i=1}^{k} N_i}$$

F : has F distribution with df_B and df_w degree of freedom

Example

- Is there a significant difference between the averages of the numerical variable (Humidity) in the three categories of the categorical variable (Outlook)?

Outlook	Humidity				
overcast	86	65	90	75	
rainy	96	80	70	80	91
sunny	85	90	95	70	70

Example

Outlook	Count	Mean	Variance
overcast	4	79.0	127.3
rainy	5	83.4	104.8
sunny	5	82.0	132.5

Source of Variation	Sum of Squares	Degree of freedom	Mean Square	F Value	Probability
Between Groups	44.0	2	22.0	0.182	0.836
Within Groups	1331.2	11	121.0		
Total	1375.2	13			

There is <u>no significant difference</u> between the averages of Humidity in the three categories of Outlook.

Practice